Jiin Im

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Research Interests

My research interests center on understanding the dynamic 3D world. I am interested in how visual systems can reason about the structure of a scene over time and learn representations that connect observations across time. Looking forward, I hope to study 4D scene understanding and representations that capture meaningful changes in the physical world.

Education

Hanyang University Seoul, South Korea
M.S. in Electronic Engineering Mar 2024 – Feb 2026
- GPA: 4.31 / 4.50
- Advisor: Prof. Je Hyeong Hong

Hanyang University Seoul, South Korea
B.S. in Electronic Engineering Mar 2020 – Feb 2024
- GPA: 4.15 / 4.50 (Major GPA: 4.22 / 4.50)
- *Summa Cum Laude*

Publications

Chebyshev Differential Flows for Shape Matching and Interpolation with Endpoint Guidance
Jiin Im, Sisung Liu, Je Hyeong Hong. *Advances in Neural Information Processing Systems (NeurIPS), 2026.*

Shape-of-You: Fused Gromov-Wasserstein Optimal Transport for Semantic Correspondence in-the-Wild [arXiv]
Jiin Im, Sisung Liu, Je Hyeong Hong. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.* Presented at KCCV 2026, KAIA 2026.

FUN-AD: Fully Unsupervised Learning for Anomaly Detection with Noisy Training Data [arXiv]
Jiin Im*, Yongho Son*, Je Hyeong Hong. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.*
* Equal contribution.

Research Experience

Spatial AI Lab, Hanyang University Seoul, South Korea
Researcher Mar 2024 – Present
Advisor: Prof. Je Hyeong Hong
- Investigating monocular dynamic 3D reconstruction with a focus on recovering temporally consistent explicit meshes and surface motion correspondence. Exploring how dynamic Gaussian representations can be coupled with explicit surface geometry to maintain geometric and temporal consistency.
- Found that temporal rigidity priors in joint shape matching and interpolation can regularize the trajectory at the expense of endpoint fitting. Introduced Chebyshev Jacobian trajectories with endpoint-null regularization to separate path regularity from endpoint alignment, leading to *Chebyshev Differential Flows* (NeurIPS 2026).
- Observed that local pseudo-label matching for semantic correspondence ignores global structure and remains ambiguous under 2D appearance cues. Reformulated matching as a 3D geometry-guided Fused Gromov-Wasserstein problem, leading to *Shape-of-You* (CVPR 2026).
- Addressed the dependence of standard one-class anomaly detection on clean normal training data, which makes it vulnerable to contamination. Leveraged pairwise feature statistics for fully unsupervised anomaly detection, leading to *FUN-AD* (WACV 2025). *Supported by Samsung Electronics DS and Korea Evaluation Institute of Industrial Technology (KEIT).*
- Observed that certain feature manifolds preserve matching performance while substantially limiting reconstruction network learnability. Currently investigate the underlying failure of reconstruction decoders theoretically and develop

corresponding attack scenarios, with initial results confirming successful attacks. *Supported by the National Research Foundation of Korea (NRF).*

- Built a pipeline for registering deformed partial point clouds from single-view depth observations to undeformed CAD models. Used learned 3D correspondences and RANSAC to obtain rigid initialization, selecting among alignment hypotheses by Chamfer distance before non-rigid registration. *Supported by CMES.*

Korea Institute of Science and Technology (KIST) Seoul, South Korea
Undergraduate Researcher Jun 2022 – Aug 2022
Advisor: Dr. Yoonseob Lim

- Analyzed nonverbal behavior in interview videos using optical flow and developed a MATLAB pipeline for real-time EEG acquisition and analysis of responses to auditory stimuli.

Patents

[PT1] Method for Anomaly Detection Based on Unsupervised Learning and System Thereof I. Jiin Im, Je Hyeong Hong. *KR Application No. 10-2024-0069569.*

[PT2] Method for Anomaly Detection Based on Unsupervised Learning and System Thereof II. Jiin Im, Yongho Son, Je Hyeong Hong. *KR Application No. 10-2024-0140579.*

Awards & Honors

BK21 Best Paper Award, Hanyang University 2026
Selected as the sole recipient from the lab.

BK21 IC-PBL+ Research Excellence Award, Hanyang University 2025
Placed among the top three of approximately 20 teams in a graduate research presentation course.

HY-IN Scholarship (Graduate), Hanyang University 2024
Awarded approximately KRW 20 million (approx. USD 14,000).

LG Electronics Paper Award, IEIE Summer Annual Conference 2023
One of five papers recognized with the award, for work developed from my undergraduate thesis; approximately 850 papers were presented.

Honorable Mention, Graduation Project Competition, Hanyang University 2023
Selected as a top-8 finalist project among approximately 131 graduating students in Electronic Engineering.

Hanyang Learning Pacemaker Scholarship, Hanyang University 2021–2023
Four-time recipient through semester-long collaborative learning in four-member teams.

Department of Electronic Engineering Scholarship (Undergraduate), Hanyang University 2020
Full tuition scholarship totaling approximately KRW 19 million (approx. USD 13,300).

Academic Service

Reviewer
NeurIPS 2026; 3DV 2026, 2027
External Reviewer: CVPR 2024, 2025; ICCV 2025; ICASSP 2025; AAAI 2024; BMVC 2024; ICIP 2023

Teaching Assistant, Hanyang University
APA4055, Introduction to Artificial Intelligence (Spring 2024, Spring 2025); ELE4092, Introduction to Computer Vision (Fall 2024); MAT2003, Linear Algebra (Spring 2024)

Skills

Programming Languages | Python, C, C++, Java, MATLAB
Research Tools | PyTorch, NumPy, OpenCV, Open3D, Blender, Docker, Linux, Git